“Patients Are Having Toxicity and Effectiveness Concerns with Pluvicto” (Brian McCloskey) [#55]

Brian McCloskey and Brad Power
May 3, 2023

“While 6 of 7 Pluvicto patients interviewed would recommend Pluvicto to a friend, that recommendation wasn’t enthusiastic, with an average satisfaction rating of 3 out of 5. With an average response duration of 6.5 months, patients found the treatment fell short of expectations.” – Brian McCloskey

“Heterogeneity of cancer limits the effectiveness of Pluvicto. Five of seven patients saw PSA declines during their treatment, but resistance occurred usually after the fourth round. Only two patients received the six full rounds of treatment.” - Brian McCloskey

Background

Pluvicto (a drug which kills cancer cells by binding to the Prostate-Specific Membrane Antigen on prostate cancer cells with a radioactive molecule, lutetium) has become a very popular treatment option for advanced, metastatic prostate cancer patients. It was approved by the FDA and came on the market in March 2022. It is a frequently prescribed treatment option and has been hard to get due to more demand than supply. In March 2023, the manufacturer (Novartis) said that the drug "is in such short supply that it cannot allow further supply to new patients until it can produce more of the drug. The company said it is working to produce enough doses to treat existing patients."

The intent of this discussion was to understand Pluvicto from patients who have experienced it, and help prospective Pluvicto patients make their treatment decisions. Some of our Prostate Cancer Lab members have been early adopters of Pluvicto, and in this discussion they raised some questions about the efficacy of the treatment and potential toxicity.

In this interactive discussion, we interviewed seven Prostate Cancer Lab patients who have been on Pluvicto to explore their experiences. Six of the seven have completed and changed to new treatments, while one patient just completed his third round of treatment and will proceed to his fourth round.

Summary Findings

- *Patient Profile (seven patients)*
    - These are all advanced prostate cancer patients who have seen, on average, four systemic therapies.
    - All but one has been on first and second-line hormone therapy.
    - All but one has been on chemotherapy.
    - All patients have bone metastases, with the exception of one who has lymph node metastases only.
- *Screening*

    - Pylarify-only: Plylarify (PSMA-PET, a scan that lights up cells with prostate-specific membrane antigen) SUVs (Standard Uptake Values) are the primary means to determine Pluvicto candidacy. SUVs ranged from 7-24.
    - Pylarify + Axumin PET: One patient treated at MD Anderson used a combination of Pylarify and Axumin PET to assess the concordance of PSMA-avid disease.
    - Genomics/RNAseq: No Pluvicto patient used RNA sequencing or other genomic insights to assess candidacy. PSMA gene expression relative to other prostate cancer patients could help determine patient fit with Pluvicto treatment. However, one Prostate Cancer Lab patient has used his PSMA RNA gene expression relative to a 1,000 prostate cancer cohort to deprioritize Pluvicto as a treatment option.
- *Treatment Response*
    - Durability: With an average response duration of 6.5 months, patients found the treatment fell short of durability expectations.
    - PSA response: Five of the seven patients saw a PSA decline through the third round. However, those patients saw PSA increases after the fourth round and discontinued. Two patients continued through the prescribed sixth round.
        - Note: Two patients experienced immediate PSA increases.
- Side Effects
    - Most common: Fatigue was the most common side effect. One patient felt that exercise was important to minimize fatigue.
    - Most serious:
        - One patient lost his salivary glands and his appetite. He was very sick from treatment.
        - One patient experienced myelosuppression, but it is not clear that Pluvicto was the driver.
- Follow-up to Pluvicto Treatment
    - Three patients switched from Pluvicto to a chemotherapy
    - Two patients moved to Bipolar Androgen Therapy (BAT)
    - One patient pursued spot radiation

Patient Data Table

Pluvicto Patient Interviews 20230503

(Please click on the link above or see the table at the end of the transcript.)

Additional Context

The manufacturer of Pluvicto is Novartis. Pluvicto is a key launch for Novartis and part of the company’s broader interest in radiotherapy. Banking on the potential to move into earlier prostate cancer treatment, Novartis put Pluvicto’s peak sales projection above $2 billion. Novartis aims to have a 250,000-dose annual capacity for Pluvicto in 2024.
From FiercePharma, "250,000 doses would translate into $8 billion of revenue at Pluvicto’s current U.S. price. Doctors’ experience thus far shows that Pluvicto has yet to reach its full potential in post-taxane mCRPC. In the clinical trial used for its approval, the drug showed it could reduce the risk of death by 38% when added to best standard-of-care in PSMA-positive

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mCRPC patients who had tried an androgen receptor inhibitor and at least one taxane-based chemotherapy regimen. And Novartis just reported positive pre-taxane mCRPC data from the phase 3 PSMAfore trial. If approved in that setting, Pluvicto’s eligible patient pool could expand from 27,000 to 42,000 patients... The company is accumulating more data at the FDA’s request, with a plan to file for an approval in the second half of this year. In addition, Novartis is testing Pluvicto in metastatic hormone-sensitive prostate cancer in the PSMAddition trial, which has a primary completion date in 2024."

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Meeting Notes

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SUMMARY KEYWORDS

docetaxel, psa, treatment, patients, russ, side effects, scan, neuroendocrine, prior, bone, infusions, suv, androgens, chemotherapy, salivary glands, bat, expression, started, fatigue, chemo

SPEAKERS

Robert Gurmankin, Brian McCloskey, Brad Power, Russ Hollyer, Chad Magnussen, Ken Anderson, Mike Yancey, Amit Gattani, Emma Shtivelman, Rick Stanton, Mark Lanctot

Brian McCloskey

The reason why we're having this discussion today is that a couple of weeks ago, there was some discussion around Pluvicto. It centered around its effectiveness and the side effects that you had. I know personally, as well as talking to many of you, that depending upon where you are in your journey, it seems to be the go-to drug treatment. What we think would be helpful is to collect any information that we have from those who have gone through Pluvicto, to understand its effectiveness and its side effects. We can use that obviously to educate other patients that come into the Prostate Cancer Lab.

To that end, I put together a few unscientific questions. I'm going to read them just so that you all have a sense of what they are, and I'm going to go across the panel of patients who I know have been on Pluvicto. I'll prompt you through the questions, but let me give you a summary.

- It would be nice to know what your prior treatment is.
- For many of you, I know when you were diagnosed. If you are new, I may not know your diagnosis date and that would be helpful because that tells me where you are on your journey.
- What kind of screening did you have? Did they just look at Standard Uptake Values, SUVs? Did they look at PSMA expression, or were there other characteristics of your cancer that came to light?
- If you did start it, when did you start it?
- What was your starting PSA?
- How many rounds did you go through? Not everyone goes through all six rounds.
- What was the end date?
- What was your PSA Nadir through that process?
- Did you have any side effects?
- I'd like a very unscientific rating on the drug, such as, 1 being completely dissatisfied and 5 being completely satisfied.
- Would you recommend it to a friend?
- What's your post Pluvicto treatment, and any general commentary?

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That's quite a few questions, so we should probably try to cap it at roughly seven minutes or so per patient. I will start off with Amit, if you can give me your perception of Pluvicto starting off with what your prior treatment was. Just your immediate prior treatment, not your entire lineage but just the one prior to it.

Amit Gattani 04:27

My immediately prior treatment was chemotherapy. I was on chemotherapy before that and I was kind of refractory to that. We tried docetaxel again while waiting for Pluvicto to become available. I started on Pluvicto around August of 2022. I got three doses every six weeks. My PSA in that phase actually increased from around 284 to the 350 range over that period of time. My last dose was in December of 2022. The side effects were part of the reason we had to stop at three doses due to myelosuppression. All three cell lines were starting to do poorly at that time. Prior to that I had gone through VMAT radiation. It's hard to say that it was only Pluvicto because VMAT radiation also reduced the bone marrow capacity. Was it Pluvicto that led me down myelosuppression? It wasn't a standalone variable, and we should keep that in mind. But the reason I had to stop after three doses was that my cell lines were too low and not safe to give me Pluvicto anymore. And in that time period, we did not see any meaningful, positive result from Pluvicto either from a PSA perspective. I have had these PSMA scans in the past and my disease seemed pretty highly PSMA avid. SUV values were pretty high too. I was very eager to get this treatment, actually, because it's a targeted treatment compared to everything else. We were all very hopeful that this was a better targeted treatment approach. So PSMA scans with high SUVs indicated this was a good approach to try. With respect to the results, I've seen data that PSA is not the best indicator being on Pluvicto. I don't have a great answer that Pluvicto helped me or not, but I'll share some information. The PSMA scan before that had a high concentration PSMA indicated a high tumor volume. PSMA post Pluvicto did show some diffusion of those areas. So presumably certain tumors did benefit from having Pluvicto. Now again, that's very qualitative information. Radiology hasn't given me any measured information. I've been on multiple things at the same time so it's hard to isolate things. But we did see some dispersion happen. Some of the spots were not as dark. Now one other side note is that during this time, the disease moved from being just in the bone to bone marrow. Again, probably nothing to do with Pluvicto, but it happened during the same period of time that my disease moved from being just in the bones and bone mets to actually getting into the bone marrow.

Brian McCloskey 08:50

Super helpful. So I think you've covered everything except your post-Pluvicto treatment. Can you tell me what that was?

Amit Gattani 09:01

My post-Pluvicto treatment is I'm on chemo again now. A carboplatin plus Cabazitaxel.

Rick Stanton 09:13

Was that directed by any molecular evidence?

Amit Gattani 09:24

Rick, what was directed by molecular evidence?

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Rick Stanton 09:27

Do you have any bone biopsies that might provide molecular direction? I'm looking for mProbe.

Amit Gattani 09:47

The chemo is driven by the fact that I now have neurondocrine differentiation. The chemo is all driven by my current state of neuroendocrine plus adenocarcinoma. With all the myelosuppression and disease being in the bone marrow, my body’s ability to take treatments is very marginal.

Rick Stanton 10:14

Just curious how neuroendocrine was determined.

Amit Gattani 10:19

Through a prostate tissue biopsy. At the end of February, the prostate tissue biopsy showed that I have neuroendocrine differentiation, bone disease, and in the bone marrow...many, many things changed.

Rick Stanton 10:38

I'm heading the same way. What type of analysis from the biopsy determined neuroendocrine?

Amit Gattani 10:45

It was a basic pathology at the hospital that determined neuroendocrine. It has been validated multiple times. I had a TURP surgery at the end of February, so they were able to get some new prostate tissue because of the procedure. From that, the basic pathology and biopsy revealed a neuroendocrine component to it.

Brian McCloskey 11:23

Amit, thank you. I'm going to stick to the rigor here. Maybe these questions are kind of marketing only, but I'm going to go with it and see how this turns out. If you were to rate this drug, on a scale of one to five, from completely dissatisfied as a one to completely satisfied as a five, what would you rate this drug?

Amit Gattani 11:51

I would put it in the middle. I mean, again, I believe in targeted treatments. And I think again, everybody may respond differently. So I would not dissuade anybody from considering it seriously. Results and mileage will vary.

Brian McCloskey 12:09

Good point. Thank you. I'm going to move on to Chad. Chad, can you tell us about your experience? The first question is what was your prior treatment, just your immediate prior treatment?

Chad Magnussen 12:30

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Obviously ADT, and then I had docetaxel chemotherapy followed by a carboplatin chemotherapy.

Brian McCloskey 12:40

What was the drug that immediately preceded Pluvicto?

Chad Magnussen 12:47

Carboplatinum.

Brian McCloskey 12:50

Okay. Got it. And did you do that with cabazitaxel or just straight carbo?

Chad Magnussen 12:56

With docetaxel.

Brian McCloskey 13:04

From a screening perspective, what did you do? PSMA-PET? Did they look at SUVs? Did you do anything else to determine whether or not you are a good fit?

Chad Magnussen 13:13

Yeah, so we did PSMA-PET and looked at the SUVs. There were a few that were similar and were high. My score was a two. Mayo Clinic rates three as a high, two is a medium, one is a low and zero is a No. So basically I was at two, so I could qualify. There were five tumors that they were monitoring. I started this in April. I was patient number 90 at Mayo Clinic, right after it was approved. I just finished round six. When I started, I had a PSA of 0.81. I had a flare that went to 1.4. It dropped pretty rapidly after that and went down to 0.77. But then now, for this last one, I am now 0.93. So the roller coaster of PSA. SUV values on the five tumors - there were two tumors that had a very low SUV of like 1.5 in a week after treatment. I did just a spec, low quality scan, and it appeared those two are gone and they're not monitored anymore. For the three larger ones, the SUV values went from in the elevens to the sevens. If two are gone, obviously, that's a good thing. And then there was some effect on the SUV value. I have seven choline scans scheduled the first week in June to really tell what this has done. So as of right now, I'm not sure what's going on. I was very sick. I would put this on par with carboplatin. I was very, very sick. Loss of appetite for a good 10 days. But the biggest thing for me is my salivary glands are completely gone. I mean, I have to drink just to talk. That's been a tough one for me. I wake up in the night and my tongue is stuck in my teeth. And it's just been a really tough go for me.

Brian McCloskey 16:22

Did they tell you that you were for some reason at high risk for that. I mean, it's a risk for all Pluvicto patients, but yours seems to be more severe.

Chad Magnussen 16:37

About the fourth one, when I started talking about it, they started mentioning how it shows up on the low quality scan and you could see the jaw light up so obviously it's going to that.

Brian McCloskey 17:00

Any other side effects?

Chad Magnussen 17:01

So far my blood numbers stayed the same. They were all pretty low coming off carboplatin, but they all stayed the same. For example my hemoglobins, I started at 12.9 and now it's 12.1. It just all stayed the same, so it didn't affect my blood numbers too badly. I was working really hard to keep them up with some supplements and juicing and all that. I think that helped. But as far as that goes, just the immediate sickness for about 10 days afterwards was really rough for me.

Brian McCloskey 17:38

Do you know what you're contemplating for your next treatment?

Chad Magnussen 17:50

Immediately we're going to do some spot radiation. I was worried that I had too many spots for spot radiation. And now if these two are gone, I think I have three to spot radiate so I think that's next for me followed by Abiraterone and progesterone.

Brian McCloskey 18:09

Got it. On a scale of one to five, how would you rate the drug? One is completely dissatisfied. Five is satisfied, or completely satisfied.

Chad Magnussen 18:20

I was pretty excited for this as I was obviously one of the first ones on it. I don't think it's paid off. It wasn't what I was hoping for, but obviously the scan I have coming in June is really going to tell the story. Right now I'm in the middle.

Brian McCloskey 18:48

Would you recommend it to a friend? It sounds kind of corny, but we're just gonna go with it for now.

Chad Magnussen 18:55

I would do three of them and get a scan in the middle and really monitor what's going on. I think they really need to come up with a way to really see if this thing's working or not.

Brian McCloskey 19:10

So the jury's out? Is that a “no” or “yes”?

Chad Magnussen 19:15

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Yeah, I think I would. Yeah, I mean, obviously, you want to look at your PSMA expression score. If you're a three, obviously I would do it. Obviously not if you're a one. I think I would have monitored that pretty closely.

Robert Gurmankin 19:57

I guess this question really is for anyone who has a dry mouth. Did they give you any suggestions as to what to do? I'm a retired dentist, I'll post on the Cancer Patient Lab discusion forum some ideas about dry mouth and how to make it less bad for anyone who has had Pluvicto dry mouth. Even people on Keytruda are running into that, too.

Brian McCloskey 20:29

Do you have any immediate recommendations, Robert?

Robert Gurmankin 20:34

I mean immediately, sipping a lot of water. There's a product line called Biotene, and they make everything from mouthwash to toothpaste to a gel that you can rub in your mouth before going to bed. I had patients find that that helps keep their lips from getting stuck to their teeth or their tongue stuck. You have to be careful about sugar. When you don't have enough saliva, you're really very prone to cavities.

Brian McCloskey 21:11

Can you take any of these treatments, pre-Pluvicto to prep the system to better handle it?

Robert Gurmankin 21:21

No. They're all going to be kind of palliative. There's no good substitute for saliva. All these do is just help to try to keep the mouth from getting too acidic and to just add lubrication. But again, I'll write something up and put it in Circle. The other thing is, you want to avoid alcohol. That just dries you out. Even if you don't have a dry mouth.

Rick Stanton 22:06

Chad, can you put in the chat or on Circle any supplements for raising your blood counts because I think that's a concern for everyone. I don't know what to do. I think I'm taking zinc or something. I have no idea. But I'm not smart on that stuff. Chad, if you could help me understand what maybe I could do to get myself in better shape that would be great.

Brian McCloskey 22:41

Anybody who has any insights on that, please chime in and post them on Circle.

Mike Yancey 22:51

I asked my doctor about raising blood counts due to another reason other than Pluvicto. He basically said there's not a whole lot you can do other than Neulasta. I tried to crank up my white blood cell count, but he said there's not a lot you can do with your blood count. The body just has to recover.

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Brian McCloskey 23:18

Mike, I know you've been on Pluvicto. Let's talk a little bit about your experience. So what was your treatment that you had immediately prior to Pluvicto?

Mike Yancey 23:30

My immediatelty prior treatment was docetaxel, which I finished in November of 2021.

Brian McCloskey 23:35

Okay, and then for your screening you did a PSMA-PET, Pylarify, and you looked at your SUV values?

Mike Yancey 23:44

Yeah, sure did. I had mets everywhere, but most of the SUV values go back to the original PSMA scan. My prostate, for example, had an SUV of 42, and then I had several others that were 24, 20, 31, and 18. So I had pretty strong SUVs going into this.

Brian McCloskey 24:06

Those are pretty high. I'm not super familiar with everybody's scores, but they're certainly higher than mine. I think mine might have been 16, or actually 11 was the highest. I'm hoping that these SUVs are standardized. Does anybody happen to know if they are standardized? You would think standardized unit value would be standardized but that might be something that we can look into afterwards. When did you start Pluvicto?

Mike Yancey 24:38

My first round was August 9. At that point in time, remembering that I don't put out that much PSA. But nevertheless my PSA at that point in time was 1.3.

Brian McCloskey 24:49

Okay, and did you get through all six rounds?

Mike Yancey 24:53

No, I just got through four rounds. We finished up the last one on December 15. That was number four and at that point in time my PSA had dropped down to 0.26. And then of course I would have scheduled for another Pluvicto at the end of January, but due to some lesions on my spine that didn't put out PSMA, we had to switch gears at that point in time and start utilizing chemo, etoposide, and carboplatin.

Brian McCloskey 25:26

Did you have side effects? There's been some discussion about fatigue, salivary gland issues, and loss of appetite. Any of that for you?

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Mike Yancey 25:42

No. I had no side effects other than maybe for the first day and a half, a little bit of fatigue. I've learned with all these treatments that if you force yourself to get out and exercise, you're going to overcome the fatigue. I'm always busy, active, etc., so I can say that the fatigue was minimal to me with the Pluvicto. I just had a lot of fun playing with the Geiger counter.

Brian McCloskey 26:08

If you were to rate the drug from 1-5, 1 being completely dissatisfied, five completely satisfied. Where would you put it?

Mike Yancey 26:20

I'll put it at a three only because even though my oncologist said it worked very, very well, I had my last dose 12/15, and I would have been ready for another dose in January. The number of SUVs has started increasing, and now I'm having a lot of issues with pain. We're evaluating what to do, etc., but my durability was three to four months, pretty much standard from what you've seen before.

Russ Hollyer 26:52

You mentioned your post Pluvicto treatment, but can you tell me that again? Was that etoposide?

Mike Yancey 26:56

Etoposide and carboplatin is my last chemo treatment, my first chemo treatment back in 2021 was docetaxel.

Brian McCloskey 27:30

Would you recommend Pluvicto to other prostate cancer patients?

Mike Yancey 27:36

I would at this point due to the fact that I had very little side effects, and it did work. I think it's a toss up as to how durable it's going to be. I think that's the big open question right now.

Brian McCloskey 27:53

I think we got everything. Okay, so, Mark Lanctot. You have experienced this, and we'd love to hear about your experience. What was your treatment immediately prior to this?

Mark Lanctot 28:16

I was on Lupron and Zytiga with prednisone for about 18 months with a slowly rising PSA. I went through a trial at Dana Farber. I contacted the doctor, Dr. Beltran, directly myself. I gained entrance in that fashion. They did all sorts of scans. Now when you do a trial, they don't really share the information with the patient. It's sort of private for Novartis. But talking to the lab people, they were actually kind of excited for me because I had a very high PSMA expression, over 95%. They told me I was the first one selected to go through. I completed all six infusions.

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Fatigue is the only side effect, no salivary glands problems. My PSA after I think the second or third infusion went from five to 0.19. I had a very good initial response, and then after about the fourth infusion, my PSA started going back up. I should also mention that I had L3, L4 involvement, and a groin. I had soft tissue and bone tissue metastases involved. In the groin the Pluvicto pretty much took care of all that, which is interesting. I did go for radiation right after treatment. I had another hotspot that developed on my tailbone, and that has been taken care of. The follow-up to Pluvicto is that I'm on BAT now and responding well. I give it a three. It all depends on each individual person. They didn't share much of the preliminary information with me so it's hard for me to say where I really fit into the whole scheme of things. With Pluvicto, I will also reiterate, exercise is huge. I started in January, so I wasn't as active as I normally would be for the first couple of months. Once I started going out hiking, the effects of it were diminished quite a bit.

Brian McCloskey 30:55

Can we just jump back real quick? In terms of the screening process, Dana Farber said that you ranked in the 95th percentile for PSA expression. Did I get that right?

Mark Lanctot 31:05

That wasn't officially said to me. That was the people in the lab who told me that when I was talking with them. It didn't come from the doctor who was running the trial, Dr. Beltran.

Brian McCloskey 31:21

Did they look at your SUV values? I think there's probably two different screens that could help to determine whether or not you're going to be responsive. One is through your Pylarify SUVs and then the second would be through, for example, RNAseq where you're looking at your PSMA expressions. Did they look at that?

Mark Lanctot 31:42

Like I said, I went through lots of tests and lots of scans, but that information was not shared with me because it was in a trial study. It's sort of a black hole as far as information that you're looking for, unfortunately. But I'm chemo naive. I had my prostatectomy almost 16 years ago. I was only on hormone therapy for 18 months. And I went from that right to Pluvicto.

Brian McCloskey 32:17

When did you start Pluvicto and when did you end?

Mark Lanctot 32:22

I would say January of last year. There was an interruption in production. I don't know if you could say I had COVID, I had got the flu. So that post, it was almost a 10-month period for this for a total of six infusions.

Brian McCloskey 32:42

Okay, so that would take you out to October 2022?

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Mark Lanctot 32:48

Yes.

Brian McCloskey 33:05

Would you recommend this to a friend?

Mark Lanctot 33:08

I guess. The only thing I would add, everyone seems to be in the three range. I would agree with that. I guess it depends on the situation of the patient too and with the testing. It sort of bought me at least a year's time, and then I met Russ, and I moved into BAT. It was good for me. I did respond. It just wasn't durable.

Rick Stanton 33:44

What ADT were you on while you were taking BAT?

Mark Lanctot 33:49

Okay, I should have mentioned that. They dropped Zytiga and they kept me on Lupron.

Rick Stanton 33:57

That's interesting. Probably because it was so early days because nowadays since the VISION trial, most people were on some type of ADT in addition to Lupron. I was on Zytiga. What's interesting also is that your next therapy was BAT, and I would think that Zytiga would queue up.

Mark Lanctot 34:32

Yes, I agree. I'm doing this outside of the standard of care. We're now getting off into a different topic, but I am sort of in communication with Sam Denmeade at John Hopkins, with Beltran at Dana Farber, and my regular MO as well. They're all monitoring me. I have a spreadsheet I send to him, and when they feed back to me, they thank me. I'm probably more of a curiosity, as Russ talks about. We're using propionate and most of the medical community is using cypionate.

Brian McCloskey 35:13

That is going to be a really good discussion, which is next week, which Russ is going to hold. Mark, I think that your experience is going to be really helpful for people. And of course, Rick, you're just starting it. I know that we've got Ken on as well.

Ken Anderson 35:50

I am one of those Pluvicto guys. I did it as soon as I could find someone locally to do it. Of course, in the beginning, I was looking under that expanded access. I was looking for someone locally versus traveling to Germany or Australia and was really close to pulling the trigger and paying 45 grand for Pluvicto, which would have made absolutely no sense for me. Thank God, I

didn't have to do it. I found a local guy in Phoenix who was willing. I was the first guy in his last clinic that did it with him.

Brian McCloskey 36:37

What was your treatment immediately prior to Pluvicto? I think it was Docetaxel. Is that right?

Ken Anderson 36:47

Yes. And I have comments in regard to chemotherapy prior to Pluvicto, but you can save that for later. My round of chemo ended in March 2022, and I started Pluvicto in April. My PSA at that time was 96. I did four cycles and never really got a response from it. I mean, I can honestly say that prior to starting Pluvicto, Paul Corn of MD Anderson's comment was, "an unremarkable therapy" – “unremarkable” because of its durability. He said some guys got responses, but it wasn't something that was durable. So most of the guys that I know that have done it, that hasn't been the case. I would imagine that's the reason for the three. Most people are grading it a three. Pretty well tolerated. I didn't really have any side effects and never held my PSA.

Brian McCloskey 37:59

What was your nadir for your PSA, when you're on it?

Ken Anderson 38:03

Went from 96 to 177.

Brian McCloskey 38:12

Then what was your end date? When did you finish, after four cycles?

Ken Anderson 38:17

July actually. I did another round of chemotherapy right after that because my PSA was rising. I did cabazitaxel. Please tee up some questions so I know what to answer.

Brian McCloskey 38:42

Cabazitaxel was the drug that you went to immediately after Pluvicto, right?

… Yep. Right.

Brian McCloskey 38:51

In terms of your screening, did you do Pylarify?

Ken Anderson 38:55

I had an Axumin PET scan a month prior to deciding to go on Pluvicto, and then prior to going on Pluvicto, I did a PSMA PET scan. Paul did a comparison between those two scans and found that they all showed up on the PSMA PET scan and the Axumin scan. I thought I was a

really good candidate for it. His comment was there's some evidence today that prior chemotherapy makes the treatment less effective, even though it's a requirement. He said that people that did chemotherapy prior to starting Pluvicto got less response. He said it wasn't really effective in bone metastases.

Brian McCloskey 39:47

If it's not effective in bone, did he talk about doing radium223?

Ken Anderson 39:54

That's not something he's really keen on, but it's still on my list. I know one guy from China that was on radium223 for four years. And it was super effective and held his PSA and his cancer. Yeah. Rare.

Brian McCloskey

Would you recommend it?

Ken Anderson

That's a tough one. Of course, as we bang through all these treatments, it's one that's in our bag, and you have to do it. I can't not recommend it. I probably wouldn't give it a three though I'd probably give it a two. And it wasn't effective for me. So of course, I'm not really keen on it. But I'd probably do it again, knowing what I know. I tried to do it with less chemotherapy upfront. I don't know whether that would matter or not. I did 27 rounds of Docetaxel prior to doing Pluvicto. If someone doing six rounds is not effective, then someone doing 27 is probably less effective.

Brian McCloskey 41:22

Your story is remarkable regarding the number of rounds of chemo that you had. I think you've set a record.

Ken Anderson 41:32

Well, that's the next topic we're going to come to next week is bipolar androgen therapy. I have a lot to say about that as well.

Ken Anderson 41:51

I'm wondering, Rick, if you're getting any pain?

Rick Stanton 41:55

I did. My spine. I've never had back pain in my life and my spine is hurting. The ribs are tender.

Ken Anderson 42:12

Muscular pain or no?

Russ Hollyer 42:15

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No. Probably inflammation.

Rick Stanton 42:20

They were a flare-up.

Ken Anderson 42:26

We had one guy at the end of last month that started BAT, had severe pain, was rushed to the ER, and ended up (because of the stress in my mind) having a brain aneurysm and passed away.

Brian McCloskey 42:47

This is Dennis?

Ken Anderson 42:48

Yeah. I was kind of pushing him to do it. So was Bryce. It was just painful, and I had asked him to get something on hand just in case he got severe pain, and he wasn't prepared for it. Bryce had pretty severe pain. It made him cry the first week. It's not without cost. Russ, I know you're going to have comments with regard to all this.

Russ Hollyer 43:26

Yeah, I do.

Ken Anderson 43:30

We probably should hold off till next week because it's everyone who wants to do this right now, and it's like anything else though. It's one of those things. It's in the bag and you almost have to do it. I mean, you bang through all the other treatments, and Pluvicto is out there. BAT is out there now. Two years ago, Paul was like, "Absolutely not. You're not doing bipolar antigen therapy." He was dead set against it. Now, he's got four or five guys at MD Anderson that are doing it. I must admit, I know lots of guys now that are doing it, and it scares the hell out of me.

Brian McCloskey 44:12

BAT is a compelling topic, and Russ has a lot to say about it, and Emmanuel Antonarakis will too. I have a decision coming up.

Russ Hollyer 44:51

I just wanted to make a quick comment that there's a lot of evidence that ADT (androgen deprivation therapy) overexpresses or causes PSMA expression to go up, and Xtandi causes it to go up. Some other ARBs might too.

Rick Stanton 45:13

Why Xtandi? It's an androgen receptor blocker?

Russ Hollyer 45:19

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It's a receptor blocker, and it's a signaling inhibitor as darolutamide, apalutamide are also, and Erleada (apalutamide). They're all doing kind of the same thing. Casodex is only a receptor blocker. The signaling is when the androgens attach to the androgen receptor and then they translocate to the cell nucleus.

Rick Stanton 45:49

I would have thought that Zitiga (abiraterone) would increase expression more than Xtandi> It just makes sense, but maybe I'm wrong.

Brian McCloskey 46:21

I can tell you that my PSMA expression went down after Zytiga.

Rick Stanton 46:34

Okay, but your AR expression is kind of what's driving the cancer battle?

Brian McCloskey 46:42

Yeah, my AR expression is still high.

Rick Stanton 46:44

So for Russ to say a blocker and a signal translocation modulation...

Russ Hollyer 46:53

There's a lot of evidence about that. When I first started looking into Pluvicto, I was hoping that androgens would overexpress PSMA. Because if they do, they would synergize well with the DNA double strand break part of it. But I found out that PSMA expression is decreased with anti-androgens. Yeah, it's one of the things where I got the disappointing news so I kept looking and looking and looking, and I kept running into Xtandi. That's all.

Brian McCloskey 47:29

I think we've got two more patients. And at least there might be a couple more. I'm going to ask Phil about his experience.

Rick Stanton 47:49

I don't think Phil's on the line. I talked to Phil and Noel for about an hour and a half yesterday.

Brian McCloskey 48:03

Let's just transition over to you, Rick. Please tell us about your experience and let's start off with what your prior treatment was.

Rick Stanton 48:16

Docetaxel. I did 11 rounds full dose, and it beat me up. My PSA went from about three to five, and it was slowly going up. But it was apparent that I should stop because I was weak. For

example, I could barely do one pushup. My arms were trembling, and I was just weakened. So the decision was to go to Pluvicto. There was a delay in getting it. I was six weeks out from being able to get my first dose of Pluvicto. Maybe a month. I was on docetaxel only. So thanks to Dr. Olson, wink wink, I started Abiraterone. My PSA was 13 at that time, and it was doubling fast. I'd say every two weeks. Suddenly I lost control. When I took all therapy away my doubling time really got short. I was 13 on Abiraterone. It went to 10 in a week. I've been doing weekly PSA monitoring which is a pain in the ass, but I was hoping to dial into the Moffitt center. I've been doing weekly PSA under Dr. Lemanne. I never got modeled. That could be my fault.

Brian McCloskey 50:13

Was it docetaxel immediately prior to or was it Abiraterone? I think it was abiraterone. When did you start Pluvicto?

Rick Stanton 50:18

It was a short pulse of abiraterone. It was only like, two weeks before I got the Pluvicto. So when I started Pluvicto, my PSA was 10, and it went down to three. But as Dr. Shen and I observed, we don't really know whether it went down to three because I had just started Abiraterone or whether it was the Pluvicto. And the way I failed was I failed at round four. It started going up from three to five to seven to 12. I doubled to 24 in a week, right before my fifth Pluvicto. If I went from 12 to like 13, I would have had the fifth Pluvicto. I had no side effects really except for a little fatigue. But no biggie. But when I went to 24, it was clear, like a sign from God, I had to move to something else. Dr. McKay highly recommended radium 223 and olaparib, which I feel is swirling down the toilet bowl to death and just putting a lot of toxicity in my system. And then there was BAT. Would I recommend it to a friend? Three. I went in hoping that I was an exceptional responder, one of those 8.2% in the VISION trial. I talked to a doctor at the Block Center who said I failed. “A lot of people fail Pluvicto in the fourth round. Join the club. And you have about a year, and it's not going to be fun." That was depressing as hell. I decided to roll the dice with BAT because that's pretty non-toxic. That's a discussion for next week. I started Pluvicto on September 30, 2022.

Brian McCloskey 52:54

You had four rounds, and you ended when? I guess, March of this year?

Rick Stanton 53:06

Around the beginning of April.

Brian McCloskey 53:13

I think I've got all that. What did you use to determine whether or not it was going to be a reasonable treatment to pursue? Did you look at Pylarify or PSMA scan?

Rick Stanton 53:28

Yeah, PSMA scan.

Brian McCloskey 53:30

And you had high values. Do you recall your SUVs?

Rick Stanton 53:35

17. My nodal metastases shrunk. In my chest, they all shrunk to about half again, I don't know whether it was Abiraterone or Pluvicto or both. But my really high increase in PSA can be attributed to new bone metastases, and it wasn't in my bone before. And that seems to be taking off. Along with some of our other observations is that soft tissue seems to respond better to Pluvicto than bone.

Brian McCloskey 54:30

That's a finding, anecdotally.

Brad Power 54:38

There was an important part of Mike Yancey's story which was that he clearly had a heterogeneous cancer population. Half of it was responding and just knocking it out, and then he had some mets that just shrunk, and then he had other mets that danced forward that weren't affected by it. I think that's a key finding is it's somewhat selective if you have a heterogeneous population.

Emma Shtivelman 55:08

In Mike's case, the lesions that didn't respond to Pluvicto were not even PSMA positive.

Rick Stanton 55:21

Mine were.

Brian McCloskey 55:39

It's good to have you back, Emma. For those of you who don't know, Emma is with Cancer Commons, and she is a rock star and an incredible researcher. She is the real deal. We're super fortunate to have her with us.

Brad Power 56:23

Just quickly since you were taking notes, can you summarize what you heard so we have that on the record because I think there are a fairly consistent set of takeaways in terms of "would you recommend this to a friend" and so on?

Brian McCloskey 56:45

Some common themes I heard are that a number of the patients were on chemotherapy just prior to Pluvicto. The majority of them are on docetaxel with a couple on carboplatin. In terms of the screening, Pylarify SUV values are the norm. It's a little hard to tell from this data what truly a high SUV value is, and I think that's been my experience, where my SUV values were lower than most of the patients that have had this. That seems to be a little bit squishy because I would have qualified for it if my creatinine levels were acceptable. I do like what Paul Corn did using the Axumin scan and PSMA-PET to look for concordance between those because not all

your cancer is going to express PSMA. Understanding what that landscape looks like, what's avid and what's not, is helpful.

In terms of the average number of rounds that folks have gone through, just eyeballing it, it's right around the four number. I think we only have one or two patients who have gotten all the way through six rounds. Nobody has done more than six rounds. I'm not even sure if you can do more than six rounds. In terms of the side effects, Chad I think has had the most significant. I think Amit, you've obviously had myelosuppression, and it looks like it's not 100% clear whether or not Pluvicto was the cause. But it did go from your bone into your bone marrow. So maybe that's a reasonable inference. Fatigue seems to be the most common side effect.

In terms of whether or not our patients are completely satisfied with this. It's kind of middle of the road. It's a three and maybe a little bit less than three. I don't think anybody was a four or five. So nobody is completely satisfied with this, and I think that the big factor here is just that it doesn't provide a durable response. It's in our arsenal.

As we talked about in terms of recommending this to a friend, everyone universally said yes, but it's qualified. You have to consider the options and know that you're probably not going to get a durable response, but it is a treatment, and you should think about it.

Those are really kind of the major themes that I saw. Anybody else wants to add any overarching themes to what they heard?

Mark Lanctot 1:00:10

While at Dana Farber, they did say that they would consider additional infusions. They did qualify for it. But it wasn't a closed door at six, at least from a Novartis point of view.

Brian McCloskey 1:00:31

It would be interesting to compare the response and this data relative to radium 223. If 85% of prostate cancer patients have bone mets, the seemingly big distinction between radium 223 and Pluvicto is that Pluvicto works not only for bone mets, but also for soft tissue. Would radium 223 be better for bone mets than Pluvicto? Not that anybody would probably subject themselves to this experiment, but is there a case to be made for using them in combination, Radium 223 for bone and Pluvicto for soft tissue? Probably not due to toxicity. But doing some sort of comparison between the two, I think would be interesting.

Ken Anderson 1:01:32

I follow Peter McAllum because they've been doing Pluvicto. They're ahead of us by five years, at minimum. People there have to pay because it's not approved by their healthcare system. But it's way cheaper. It's like 600 bucks a round in Australia for Pluvicto, and they do multiple rounds. The max that they've published, one guy's done 18 rounds. The way I think that they found to get guys to respond is they add 5% actinium. They do a little concoction. The actinium goes to bone and does a better job, and they end up getting a response if they add the 5% actinium when they don't with Pluvicto on its own.

Brian McCloskey 1:02:32

“Patients Are Having Toxicity and Effectiveness Concerns with Pluvicto” (Brian McCloskey) [#55]

Do you know what the side effects are from actinium? Because you know, Bryce did it, and it was bad.

Ken Anderson 1:02:36

I know Bryce didn't handle it very well. I didn't really get any input from what was published as far as side effects from the guys that have done multiple rounds. I can't imagine having anything other than dry mouth and fatigue.

Robert Gurmankin 1:02:54

One thing I can add with actinium is that one round can kill the salivary glands, whereas with Pluvicto it often takes several rounds.

Ken Anderson 1:03:03

That didn't happen to Bryce. He was on a clinical trial with actinium, and that didn't happen. I don't know if you could make the broad statement that one round would do it. But for some guys, maybe one round does.

Robert Gurmankin 1:03:22

I should say, often, one round will kill it.

Amit Gattani 1:03:32

All of us are tuned to the notion that none of these therapies are going to be durable, so how do you sequence them, and how do you make sure that you get the right mileage out of them? I don't know about the 5% stuff. But generally, sequencing is Pluvicto first and then actinium. If you do actinium first, then Pluvicto will not be effective. I've looked into this internationally also before it became available here, and I'm aware of very long treatment cycles of Pluvicto in Germany as well. I have also heard from Germany that somebody's had 18 cycles of Pluvicto.

Russ Hollyer 1:04:41

First, a plug for BAT. Radium 223 has a combination trial with BAT right now and is recruiting.

I was wondering if any of your MOs have done Pluvicto with a PARP inhibitor, like olaparib, or any other PARP inhibitor?

“Patients Are Having Toxicity and Effectiveness Concerns with Pluvicto” (Brian McCloskey) [#55]

Patient Data Table

Pluvicto Patient Interviews 20230503

Demographics & Select EMR Data							
	Patient 1	**Patient 2**	**Patient 3**	**Patient 4**	**Patient 5**	**Patient 6**	**Patient 7**
Age	68	66	62	53	51		57
Date of Diagnosis	11/1/2019	7/20/2021	1/1/2017	11/30/2018	5/11/2021	10/1/2007	3/29/2018
Years living with Disease	3.6	1.9	6.4	4.5	2.1	15.7	5.2
Primary Treating Hospital	UCSD	Highlands Oncology	MD Anderson	UCSF	Mayo Clinic	Dana Farber/Johns Hopkins	Mayo Clinic
Gleason	7	-	9	-	-	7	9
Met Location	Lymph node and bone	Bone	Bone	Bone	Bone	Bone	Lymph Node
DNA Alterations	CDK12, TMPRSS2-ERG	PTEN, AKT2, CCNE1, ASXL1, RB1, TP53	TP53	AR, CDK12, CCND1, CDK6	N/A	N/A	CDK4, TP53, PTEN, ATM
Key Treatments	Docetaxel, Abiraterone, Pluvicto	Docetaxel, Abiraterone, Pluvicto, Etoposide + Carboplatin	Docetaxel, Pluvicto, BAT	Radiation, Enzalutamide, Abiraterone, Pembro+177Lu, Docetaxel, Cabazitaxel, Pluvicto	Docetaxel, Abiraterone, Pluvicto	Prostatectomy, Radiation, Firmagon/Lupron, Abiraterone, Pluvicto, BAT	Radiation, Docetaxel, Carboplatin, Abiraterone, Enzalutamide
Systemic Therapies	3	4	3	6	3	4	4
Years/Systemic Therapies	1.20	0.48	2.13	0.75	0.70	3.93	1.30
Pluvicto Survey							
Immediately Preceding Treatment	Abiraterone	Docetaxel	Docetaxel	Docetaxel	Carboplatin + Docetaxel	Lupron and Zytiga	Lupron + xtandi
Screening:							
SUVs	Pylarify - SUVs 17	PSMA PET - Looked at SUV values, 24, 22, etc	Axumin scan, and PSMA PET scan and there was concordance between	Pretty high SUVs. High PSMA expression	Pretty high SUV. Mayo score was 2. 5 tumors. SUVs were low for some, 1.5. SUVs for larger ones	95% PSMA Expression	Nov. 2022, PSMA-PET. Cluster in left should, lymph nodes (19.3), T3 and right hip (16.1). SUV max

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					went from 11-7		12.6. Mayo score is a 2.
RNA PSMA expression?	No	No	No	No	No	No	No
Pluvicto Start Date	9/22/2022	8/9/2022	4/22/2023	8/22/2023	4/22/2022	1/22/2022	2/8/2023
Starting PSA	10	1.3	96	284	0.81	5	0.81
How many rounds?	4	4	4	3	6	6	3
End Date	4/3/2023	12/15/2022	7/22/2023	12/22/2023	4/23/2023	10/22/2022	In progress
PSA Nadir	3	0.26	177	350	0.77	0.19	0.25
Side Effects	A little fatigue	No side effects. A little bit of fatigue in the beginning which was mitigated by staying active.	No side effects	Myelosuppressionn	Very sick. Loss of appetite. Salivary glands are gone. Hemoglobin stayed the same. Used supplements and juicing.	Fatigue	Nothing severe. Day after some nausea. Lower jaw-receding gum line and dry mouth.
Rate Drug (1-5) 1 completely dissatisfied/5 Completely satisfied	3	3	2	3	3	3	4
Recommend to a friend	Yes	Yes	Yes	Yes	Yes	Yes - it would depend on where the patient is. Not durable	No (It's not a cure.) It's a lot to go through
Post Pluvicto Treatment	BAT	Etoposide and carboplatin	Cabazitaxel	Carboplatin + Cabazatixel	Spot radiations	BAT	In progress. Still on xtandi
Commentary	* Not sure if response was from abiraterone. *Along with some of our other observations is that soft tissue seems to respond better to Pluvicto than bone.	*Had to switch *Brad Power - "Brian, I just want to add that I think there was an important part of Mike Yancey's story which was he clearly had a heterogeneous cancer population. Half of it was responding and just knocking it out, and then he had some	Never got a response *"I did four cycles and never really got a response from it. I mean, I can honestly say that prior to starting Pluvicto, my MO at MD Andersons comment was, "an unremarkable therapy" unremarkable because of its durability. *His comment was there's some evidence	* The reason I had to stop after three doses was that my cell lines were to low and not safe to give me Pluvicto anymore, and in that time period, we did not see any meaningful, positive result from Pluvicto either from a PSA perspective. * The disease moved from being just in the bone to bone marrow. Again, probably nothing	*"I was pretty excited for this as I was obviously one of the first ones on it. I don't think it's paid off." Satisfaction dependent on patient response/condition. *Advice: I would do three of them and get a scan in the middle and really monitor what's going on. I think	With Pluvicto, I will also reiterate, exercise is huge. I started in January and so for the first couple of months, I wasn't as active as I normally would be. Once I started going out hiking, the effects of it were diminished quite a bit. *"It was good for me. I did respond. It just	Going through 6 treatments. More nausea than chemo. A lot to go through for not being a cure. On paper, they say it's great, but only 15% of patients are finishing it and it doesn't seem durable. His PSA has dropped, but imaging is showing an increase in size. It could be a scar.

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		mets that just shrunk, and then he had other mets that danced forward that weren't effected to by it. I think that's a key finding is it's somewhat selective if you have a heterogeneous population."	today that prior chemotherapy makes the treatment less effective even though it's a requirement. He said, people that did chemotherapy prior to starting Pluvicto got less response? He said it wasn't real effective in bone.	to do with Pluvicto, but it happened during the same period of time that my disease moved from being just in the bones and bone mets to actually getting into the bone marrow.	they really need to come up with a way to really see if this thing's working or not.	wasn't durable. "	
Imaging Improvement	Nodal metastases shrunk	Not	Not durable	Didn't see any meaningful response	Choline scans schedule		